SIX UNIT SQUARES CANNOT COVER A SQUARE OF SIDE GREATER THAN 2

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ABSTRACT. Let $S(n)$ be the largest side length of a Euclidean square that can be covered by n closed unit squares, allowing arbitrary rigid motions and overlaps. Equivalently, Soifer's function $\Pi(2)$ is the least number of unit squares needed to cover some square of side $2+\varepsilon$ for every $\varepsilon > 0$. We prove that $S(6) = 2$, or equivalently $\Pi(2) = 7$, resolving the conjecture stated by Dósa–Lángi–Tuza [4]. In particular, six unit squares cannot cover any square of side strictly greater than 2, while side 2 is attained by the obvious 2×2 tiling.

The argument is a finite case analysis on ownership of the centre and of the four outer midpoints, followed by a reduction to a final standard form and the elimination of all joining modes. Algebraic exclusions are discharged by exact rational identities, positive Bernstein certificates on compact boxes, interval certificates, and SMT UNSAT queries; selected geometric interfaces are formalised in Lean 4. The proof is computer-assisted: machine certificates (with `still_S`=0) and a Lean 4 obligation skeleton (no `sorry`, no extra axioms) support the narrative. We do not claim independent peer review, nor a full Lean re-proof of every metric inequality appearing in the long verification manuscript.

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1. Introduction

Covering a large square by equal smaller squares is a classical theme in discrete and metric geometry. Write

$$S(n) \;=\; \sup\bigl\{\, s>0 : \text{a closed square of side } s \text{ admits a covering by } n \text{ closed unit squares} \,\bigr\}, \tag{1}$$

where each unit square may be translated and rotated and overlaps are allowed. Soifer's function $\Pi(n)$ asks for the least number of unit squares needed to cover *some* square of side $n+\varepsilon$ for every $\varepsilon>0$; see [9, 8]. In particular

$$S(6)=2 \quad\Longleftrightarrow\quad \Pi(2)=7. \tag{2}$$

The lower bound $S(6)\ge 2$ is elementary: four axis-aligned unit squares tile $[0,2]^2$. The matching upper bound is the content of the present paper.

Januszewski proved that covering a square of side $2+\varepsilon$ requires at least six unit squares, i.e. $\Pi(2)\ge 6$ [7]. Exact small values such as $S(5)=2$ appear in the literature of Friedman–Paterson and Januszewski; see also Friedman's covering surveys [6, 5]. In January 2026, Dósa, Lángi and Tuza revisited congruent square coverings, determined several small $S(n)$ and boundary-covering variants, and stated as Conjecture 1.1 that $S(6)=2$ (equivalently $\Pi(2)=7$) [4].

Theorem 1.1 (Main theorem)**.** $S(6)=2$. *Equivalently,* $\Pi(2)=7$*: six closed unit squares cannot cover any square of side strictly greater than* 2*, while a square of side* 2 *admits such a covering.*

The proof occupies Sections 2–11. Section 12 records the logical rôles of the computational certificates. Related work is collected in Section 13.

Remark 1.2 (Accuracy tier)**.** Theorem 1.1 is asserted as a computer-assisted complete proof: internal audit (round 65), certificate inventory with `still_S`=0, and a Lean 4 obligation skeleton with closed flags for obligations 1–4. This does *not* assert that the result has undergone independent peer review, nor that every metric inequality in the long manuscript has been re-proved in Lean.

2. Definitions and standing notation

Definition 2.1. For each integer $n\ge 1$, $S(n)$ is defined by (1). Soifer's $\Pi(n)$ is the least m such that for every $\varepsilon>0$ some square of side $n+\varepsilon$ admits a covering by m unit squares.

Throughout the upper-bound argument we work with the centred target

$$T = [-\mu, \mu]^2, \qquad \mu > 1, \tag{3}$$

and derive a contradiction from the assumption that six closed unit squares cover T. (The translate $[0, 2\mu]^2$ is equivalent for all length budgets.)

Write O for the centre of T, label the four corners $1, 2, 3, 4$ counterclockwise from the lower left, and write BOTTOM, RIGHT, TOP, LEFT for the outer edges. An *outer mid* is the midpoint of an outer edge. The central horizontal and vertical target segments are denoted H and V. A *true gap* on an outer edge is a nonempty open subinterval of that edge not covered by the two adjacent corner tiles; endpoints of gaps are never free hypotheses and must be recovered from openness, convexity and closedness.

"Corner tiles abut" on an outer edge means that the two adjacent corner tiles together cover the entire edge (overlaps allowed). Otherwise the uncovered open interval is a true gap.

3. Roles, normalisation, and parallel budgets

Lemma 3.1 (Roles and midpoints)**.** *Four distinct tiles V_1, V_2, V_3, V_4 cover the four corners of T; a fifth tile C covers O; the sixth tile is denoted F. Each unit tile contains at most one outer mid. The role selection does not forbid F from containing a corner, nor a mid from lying in both a corner tile and a non-corner tile.*

Proof. Distinct target corners are at distance at least $2\mu > \sqrt{2}$, hence cannot lie in one unit tile. The distance from a corner to the centre is $\sqrt{2}\,\mu > \sqrt{2}$, so C is not a corner tile. Distinct outer mids are at least $\sqrt{2}\,\mu > \sqrt{2}$ apart. □

Lemma 3.2 (Dominated normalisation)**.** *Fix a target corner as origin with inward coordinates $X, Y \geq 0$. The unit tile containing that corner may be enlarged inside T to the form*

$$0 \leq cX + sY \leq 1, \qquad -\beta \leq -sX + cY \leq 1 - \beta, \quad c, s \geq 0,\ c^2 + s^2 = 1, \tag{4}$$

without moving any other tile and without deleting any previously covered point of T.

Proof. Among the four oriented unit normals choose one in the open first quadrant. Its original projection interval $[\alpha, \alpha + 1]$ contains 0, hence $\alpha \leq 0$. Replacing it by $[0, 1]$ does not remove any point of T with nonnegative projection. Keep the orthogonal unit interval unchanged. □

Proposition 3.3 (One-shot global normalisation)**.** *Starting from any covering of T, apply Lemma 3.2 once to each corner tile V_i, leaving C and F fixed. The resulting covering still covers T, preserves roles of C and F, and is idempotent. Thereafter classification is performed exactly once: every later local normalisation step acts as the identity on already normalised corner tiles.*

Lemma 3.4 (Two nonempty parallel chords)**.** *If a unit tile meets two parallel lines at distance $d > 1$, then $d \le \sqrt{2}$ and the sum of the two chord lengths is at most*

$$B(d) = \begin{cases} d - \sqrt{d^2 - 1}, & 1 < d \le 3/(2\sqrt{2}), \\ 2(\sqrt{2} - d), & 3/(2\sqrt{2}) \le d \le \sqrt{2}. \end{cases} \tag{5}$$

In particular $B(d) < 1$. Tangency counts as nonempty; an empty chord may not be substituted into the bound.

Proof sketch. With direction parameters $c \ge s > 0$ the transverse width is $w = c + s$. The two vertical lines lie at t and $t + d$ with $0 \le t \le w - d$. Since $d > 1 \ge c$, both chords fall in the triangular end-caps and the total length equals $(w-d)/(cs) = 2(w-d)/(w^2-1)$. Maximising over $d \le w \le \sqrt{2}$ yields the two branches of (5). The Euclidean chord-length identities matching the triangular-cap formulae are recorded in the Lean development of the parallel-budget geometry (Obligation 1). □

Further elementary budgets (grid contribution at most 2 per corner tile; $L + S + U < 2$ for a normalised corner tile; triangular area at most $1/2$) are collected as Lemmas R2–R3 in the long verification manuscript and used below without restating the proofs.

4. Top-level classification

If F also contains O, call the configuration class G5. Otherwise set

$$k \;=\; \#\{\text{outer mids belonging to } C \cup F\} \in \{0, 1, 2\}. \tag{6}$$

Then $k = 0$ is G1, $k = 1$ is G2, and $k = 2$ with adjacent (resp. opposite) mids is G4 (resp. G3). Each mid not yet supplied by $C \cup F$ chooses an adjacent corner host injectively. Reflection reduces host representatives.

Proposition 4.1 (Exhaustive branch table)**.** *Every covering of T falls into exactly one of the classes G5, G1, centre-owning G2, G4, G3, or residual G2 ($k = 1$ with the unique mid owned by F on* BOTTOM*). The residual G2 splits into a near-end host representative* $(\text{RIGHT}, \text{TOP}, \text{LEFT}) = (2, 3, 4)$ *and a far-end representative* $(2, 4, 1)$*.*

The global elimination order is: G5 for all three values of k; then G1, centre-owning G2, G4 and G3; then near-end residual G2; then reduction of the far-end representative to the final standard form; then elimination of all joining modes of that form.

5. Elimination of G5, G1, centre-G2, G3 and G4

Proposition 5.1 (G5 impossible)**.** *In any covering of T exactly one tile contains O, and O is an interior point of that tile.*

Proof outline. For G5, modules R2–R3 eliminate $k = 2$ and $k = 0$. For $k = 1$, modules R4–R7 reduce to a triple-abutment configuration with an interior second centre tile; module R8 then proves the two-corner inequality

$$L_1 + L_2 + \min(t_1, t_2) + \tfrac{4}{5}\max(t_1, t_2) \le \tfrac{14}{5} \tag{7}$$

by a Bernstein certificate on forty-five rational rectangles after mapping the full two-dimensional domain into the unit square, and compares it with the weighted three-ray bound on the interior tile to reach a contradiction in both orderings of (t_1, t_2). Thus G5 is empty. The remaining five tiles stay at positive distance from O, so a small neighbourhood of O is covered by C alone and O is interior to C. □

Proposition 5.2 (G1 impossible)**.** *If the centre tile is unique, then $C \cup F$ contains at least one outer mid.*

Proof outline. Module R9 eliminates triple abutment by a quarter-boundary budget and reduces to opposite or adjacent residual gaps. For opposite gaps, the corrigendum R10C first forces the horizontal chord of F to be strictly positive (via $g_1 + h_C > 1$ against $B(\mu) < 1$), then combines parallel-chord budgets of C and F with the corner contribution bound < 8. Adjacent gaps are eliminated in R10 §§3–6 by real endpoints and direction constraints. □

Proposition 5.3 (Centre cannot own the unique mid)**.** *If $k = 1$, the unique mid belonging to $C \cup F$ cannot belong to C.*

Proof outline. Axial propagation (R11) yields necessary outer contacts. Module R12 treats the two host representatives when only C owns the mid, covering all true-gap and triple-abutment patterns on LEFT, TOP and RIGHT. Shared ownership $C \cap F$ of the same mid is eliminated separately in R12 §10 by vertical redundancy and a true-gap triangle constraint. □

Proposition 5.4 (G4 impossible)**.** *No covering has C and F owning distinct adjacent outer mids.*

Proof outline. Normalise so that C owns the left mid and F owns the bottom mid. The three injective host representatives $(\mathrm{TOP}, \mathrm{RIGHT}) \in \{(4,3), (4,2), (3,2)\}$ are eliminated in modules R20–R23 by horizontal-sign analysis, forced suspension segments, and projection budgets on C. Shared left-mid ownership by the free corner tile is ruled out before the last representative. □

Proposition 5.5 (G3 impossible)**.** *No covering has C and F owning distinct opposite outer mids.*

Proof outline. Normalise so that C owns the left mid and F owns the right mid. Three host representatives (after reflection) are eliminated in R24–R25 by abutment/gap analysis on TOP and BOTTOM, empty/nonempty vertical chords of F, and a local exit lemma for nonempty cases. □

Corollary 5.6 (Universal centre forbids outer mids)**.** *In any covering of T the unique centre tile contains no outer mid, and the remaining non-corner tile contains some outer mid.*

Proof. Combine Propositions 5.1–5.5. □

6. Near-end residual G2

After Corollary 5.6 the only surviving class is residual G2 with the unique mid owned by F on BOTTOM.

Proposition 6.1 (Near-end impossible)**.** *The near-end host representative* $(\mathrm{RIGHT}, \mathrm{TOP}, \mathrm{LEFT}) = (2, 3, 4)$ *cannot occur.*

Proof outline. Module R18 shows that either both RIGHT and TOP abut, or exactly one of them carries a true gap owned by C. In the double-abutment case: R13 eliminates nonempty horizontal chords of F; R14–R15 force left abutment when that chord is empty and produce an interior point Q_0 of C; R15S/R16P supply the strict angular separation and height hypotheses; R16 then covers the necessary continuous domain by 1173 rational boxes with interval certificates excluding every leaf. Remaining left-gap / nonempty-F patterns are eliminated in R17–R18. Top-gap patterns fall to R19 and R26; right-gap patterns fall to R27–R28 (both rotation signs). Normalisation that fills a gap returns to an already eliminated abutment leaf. □

7. Far-end reduction to standard form

Henceforth fix the far-end representative: V_1 owns the left mid, V_2 the right mid, V_4 the top mid, and the free corner tile is $W = V_3$. The centre tile C owns no outer mid; F owns the bottom mid and not the centre.

Proposition 7.1 (Far-end standard form)**.** *Every far-end covering satisfies: the three non-bottom outer edges abut via corner tiles;* $F \subset \{y < 0\}$*;* C *avoids* BOTTOM*; and a single copy of* F *contains both closed endpoints of the bottom gap together with the lower axial tip of* C*.*

Proof outline. Step 1. Modules R27 and R29 eliminate standardised top true gaps for both rotation signs of C. Thus TOP abuts. Any lateral true gap lies too far from the bottom mid for F to reach, so lateral gaps are owned entirely by C. Since C cannot meet both LEFT and RIGHT at distance $2\mu > \sqrt{2}$, the only remaining cases are left gap, right gap, or triple abutment (corrigendum R29N).

Step 2. If $F \cap H \neq \varnothing$, R29N forces C to avoid BOTTOM; R31 eliminates triple abutment and R32–R36 eliminate the lateral-gap cases. Hence $F \cap H = \varnothing$. If some point of F had $y \geq 0$, convexity with $(0, -\mu) \in F$ would produce a point of $F \cap H$ inside T, a contradiction. Thus $F \subset \{y < 0\}$.

Step 3. Corollary 5.6 applied to an inscribed target yields axial-root caps $\leq 2 - \mu$ (module R37) without circularity. Modules R37–R39 eliminate all right and left true-gap patterns (both rotation signs, direct and bridged contacts).

Step 4. In the residual triple-abutment case, R39 gives the upper axial root $u > 38/79$, whence C avoids BOTTOM. The three anchored corner tiles

can contribute only their matching central half-axes, so the entire bottom gap and the lower tip of C are forced into the same tile F. □

8. Standard-form notation and four joining modes

Let $a, x, z \in (0,1)$ be the half-angle parameters of V_1, V_2, V_4, and let $h \in [0, \sqrt{2}-1]$ be the half-angle of the smaller axial angle of C. Set

$$b(t) = \frac{1+t^2}{1-t^2}, \quad k(t) = \frac{2t}{1-t^2}, \quad g(t) = \frac{1-t}{1+t}, \quad q(t) = \frac{t}{2-t}, \tag{8}$$

and write $\delta(t) = b(t) - 1$, $\Delta_z = \delta(z)$. The four actual axial tips of C are $L = (-l, 0)$, $R = (r, 0)$, $T_C = (0, u)$ and $B_C = (0, -d)$. Writing S_1, S_2 for the actual short legs of the bottom corner tiles and $\Sigma = S_1 + S_2$, Proposition 7.1 yields

$$\Sigma > 2 - \sqrt{2} > \tfrac{7}{12}, \qquad \Sigma > (1-d)^2, \qquad \Sigma > 2 - \frac{1}{1-d}. \tag{9}$$

Left abutment supplies the constraints

$$0 < z < a < 1, \quad a+2z < 1, \quad z < \tfrac{4}{15}, \quad 0 \le S_1 < \mathcal{F}(a,z) := b(a) - k(a)\big(2 - g(z)\big). \tag{10}$$

Lemma 8.1 (Normalised seam point in C)**.** *There is a point*

$$P_1 = (-1 + X, e) \in C, \qquad X = \frac{b(a) + S_4 - 2}{k(a) - k(z)}, \quad e = 1 - S_4 - k(z)X, \tag{11}$$

obtained from the real-scale seam $P_\mu \in C$ by a convexity argument inside the same tile C (modules R40 and R42). The construction does not rescale the covering to $\mu = 1$ and does not move C.

Let Q_2 and Q_U be the actual inward tips of V_2 and V_4 on the positive axes. Whether C contains these tips yields four exhaustive joining modes: double-direct, upper bridge by W, right bridge by W, and double bridge. If C misses a tip, the corresponding open axial gap can be filled only by the same free corner tile W, which is then forced by closedness to contain both the tip of C and the corner tip.

9. Elimination of upper bridges and negative rotations

Proposition 9.1. *In the standard form of Proposition 7.1, the free corner tile W cannot contain Q_U.*

Proof outline. Modules R44–R45 show that an upper bridge forces $z \le \tau < 2/5$ for the parameter τ of W and empties $W \cap H$, whence C contains Q_2 directly and double bridging is impossible. Modules R46 and R48 produce a real exit point $T_R \in C \cap W \cap V_2$. Small angles $h \le 1/24$ fall to R52. Positive rotations are eliminated in R53–R55 via the seam-derived bound $S_1 < q(Q)$

and the two normals of C at T_R. Negative rotations are reduced in R56–R58 to $0 < \tau < 1/6$ and then excluded in R59 by the area-forced negativity

$$G = H_F - q(x+\delta) - g(x) + k(x)(B_t - 1) < 0, \quad H_F = 2 - \frac{1}{1-\delta-\tau}, \tag{12}$$

using endpoint polynomial certificates on small/medium x and a left-root projection contradiction on large x. □

Consequently only double-direct and right-bridge modes remain, both with $C \ni Q_U$ and hence $u > 1 - z$. In the right-bridge case the horizontal chord length of W is strictly less than Δ_z, which after establishing that the relevant parameters are < 1 yields

$$S_2 < q(r + \Delta_z); \tag{13}$$

double-direct admits the stronger $S_2 < q(r)$.

Proposition 9.2. *All negative rotations in the two residual joining modes are impossible. Every surviving positive rotation satisfies* $1/10 < z < 1/5$.

Proof outline. Small-angle double-direct and right-bridge cases are eliminated in R43 and R49–R51 respectively, both forcing $h > 1/8$. Module R60 partitions the negative-rotation domain into four regions and contradicts (9) in each, paying the loss (13). Modules R61–R62 eliminate $z \geq 1/5$ and $z \leq 1/10$ for positive rotations by asymmetric triangles on the same F and concavity of the budget form, leaving the open strip $1/10 < z < 1/5$. □

10. Right bridges are impossible

Proposition 10.1. *In the residual domain of Proposition 9.2, W does not contain the right axial tip R of C. Consequently the entire right-bridge class is empty.*

Proof outline. Module R63 first obtains $r < 17/20$ from axial chords and the R/T_C projection. Assuming $W \ni R$, the tile meets both TOP and H, forcing a top leg $D > 1 - \Delta_z > 11/12$ and a half-angle

$$t < \tfrac{1}{8}, \qquad t < \tfrac{9}{7}\Delta_z. \tag{14}$$

Comparing real projections in both orientations of W yields an abscissa ξ of the contact on H satisfying

$$\xi < \Delta_z + t \quad \text{or} \quad \xi > 1 - t > \tfrac{7}{8}. \tag{15}$$

The second alternative contradicts $r < 17/20$ if the contact is R. The first, combined with (13)–(14) and left long-leg loss, produces

$$\Sigma < \Phi\Big(\frac{2z}{1+z}\Big) + q\Big(\frac{23}{7}\Delta_z\Big), \quad \Phi(w) = (1+w)\big(1 - \sqrt{w(2+w)}\big), \tag{16}$$

which is decreasing on $1/10 \leq z \leq 1/5$ and strictly less than $7/12$ at the left endpoint, contradicting (9). □

11. Double-direct positive rotations

Proposition 11.1. *No positive-rotation double-direct covering survives in the residual strip of Proposition 9.2.*

Proof outline. From (10) and $1/10 < z < 1/5$ one obtains $2/7 < a < 3/5$. Double-direct gives $S_2 < q(r)$. At the formal endpoint $S_4 = g(z)$ define

$$X_G = \frac{(1-z)\big[a^2(1+2z)-z\big]}{(a-z)(1+az)}, \qquad e_G = \frac{2az(1-a-z)}{(a-z)(1+az)}. \tag{17}$$

After verifying $a < g(h)$ one has the projection bound

$$r < R_P(h) := \delta(h) + X_G + k(h)e_G, \tag{18}$$

while the pair (T_C, R) yields $r < R_T(h) := h + z/k(h)$ for $h > 0$.

On $0 \le h \le 1/4$ one has $a < 3/5 \le g(h)$; monotonicity of (18) reduces the right-root upper bound to $h = 1/4$, after which $\mathcal{F} + q\big(R_P(1/4)\big) < 7/12$ on the full (a, z)-rectangle. On $1/4 \le h \le 1/3$, concavity of the real P_1/T_C relation forces $a > 2/5$, and four endpoint values control $\Sigma < 7/12$. On $1/3 \le h \le \sqrt{2} - 1$, a rational outer cap $29/70$ splits according as $a \ge g(h)$ or not; both subcases again force $\Sigma < 7/12$.

The fourteen deduplicated rational targets are certified by twenty-eight numerator/denominator polynomials whose one hundred fourteen strictly positive rational Bernstein coefficients, together with reconstruction identities, cover the whole continuum $[0, \sqrt{2} - 1]$ (module R64). Every angular segment contradicts (9). □

Proof of Theorem 1.1. Assume six unit squares cover T with $\mu > 1$. Proposition 4.1 places the covering in one of the top-level classes. Propositions 5.1–5.5 eliminate G5, G1, centre-owning G2, G3 and G4. Proposition 6.1 eliminates the near-end residual G2. Proposition 7.1 reduces the far-end residual G2 to the final standard form. Propositions 9.1–11.1 eliminate every joining mode, rotation sign and residual parameter domain of that form. Hence $\mu > 1$ is impossible. Combined with the elementary covering of side 2, one obtains $S(6) = 2$, and therefore $\Pi(2) = 7$ by (2). □

12. Computational certificates

The geometric narrative of Sections 3–11 is supported by three layers of evidence.

Exact rational certificates. On a polynomial over $[0, 1]^m$, the tensor Bernstein basis is nonnegative and sums to one. Nonnegative (resp. strictly positive) coefficients prove nonnegativity (resp. strict positivity). Parameter domains are mapped into rectangles by explicit rational substitutions after denominators are shown positive. Verifiers regenerate targets, rebuild basis-change identities and rational coefficients, rather than trusting a generator's claimed minimal coefficient. Module R8 additionally checks a complete prefix-tree cover; module R64 checks every polynomial on its fixed rectangles.

Continuous interval certificates. The R16 root box outwardly contains the necessary domain. Each internal node splits into children whose union recovers the parent; each leaf is excluded by exact fractional envelopes. Denominator positivity, monotonic substitution direction, and all leaves together with their boundaries are checked.
Real-algebraic UNSAT queries. Selected propositions are accompanied by QF_LRA or QF_NRA queries. Only UNSAT is accepted. Queries are classified as (E) exact certificates, (H) hand proofs with solver cross-check, or (S) dual-solver UNSAT (z3 and cvc5) not yet reduced to (E). Dual-wrapper spot checks on the M8–M11 inventory report pass on the expanded faithfulness sample shipped with the ancillary package. Finite floating-point sampling is never used as an exclusion certificate.
Lean 4 fragments. Selected interfaces—parallel-budget geometry and Euclidean chord lengths; reduction of an arbitrary covering to a manuscript-normalised model; chord and ownership bridges; M7.1 narrative↔checker branch logic and the `StandardForm` predicate; far-end PDF faithfulness and face automata; and algebraic/geometric pieces used in M8–M11 together with the seventeen living near-end leaf wirings—are formalised in the Lean 4 package `S6Core` (mathlib), building with no `sorry` and no extra axioms. The long verification manuscript remains the geometric reference for the many metric inequalities that are discharged primarily by certificates rather than by a full Lean re-proof of every inequality; ancillary files record the correspondence between manuscript modules and certificates.
Reproduction. The ancillary archive contains the Lean sources, certificate registries, checker scripts for representative rounds (including R29, R38, R42, R55-final, R64, and the R16 leaf cover), and the long verification manuscript. A minimal path is

```
cd formalization/lean/S6Core && lake build
python3 formalization/audit/g3_alignment_check.py
# spot-check ex/all/check_round{29,38,42,55_final,64}.py
```

13. Related work

Soifer's JCTA note and Cover-Up Squared writings introduced $\Pi(n)$ [9, 8]. Januszewski's Monthly note established $\Pi(2) \geq 6$ [7]. Friedman–Paterson and Friedman's surveys record constructions and small-case discussions [6, 5]. Dósa–Lángi–Tuza proved several exact small values and boundary analogues and isolated the conjecture $S(6) = 2$ [4]; Theorem 1.1 resolves that conjecture. Asymptotic packing-waste problems of Chung–Graham [2, 3] concern a different regime.

On computer-assisted proofs in geometry, we follow the AMS guidelines on independent checkability and durable availability of code and data [1].

14. Concluding remarks

Theorem 1.1 determines the first previously open value of $S(n)$ beyond the classical small cases. The same case-analysis and certificate pipeline suggests approaches to neighbouring values (e.g. refinements of $\Pi(3)$ and boundary-covering variants $S_{\mathrm{bd}}(n)$ from [4]), though those questions lie outside the present paper.

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